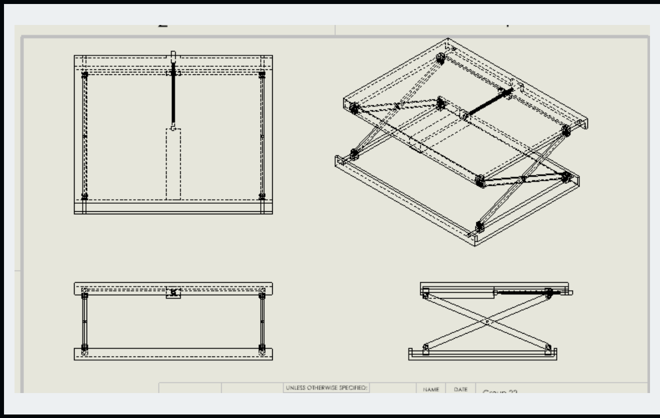
ENME 339 GROUP DESIGN

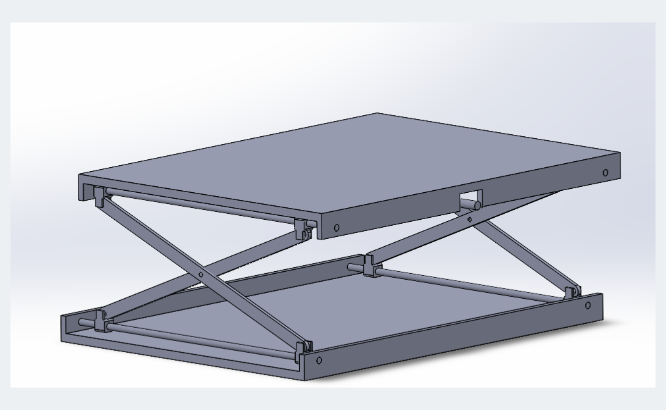
SCISSOR LIFT WORKBENCH

This was completed as part of an ENME 339 group design project. I modeled the main table structure, support bars, pivot pins, lead screw, nut block, and mate relationships for a lead-screw-driven scissor lift workbench in SolidWorks. The mechanism converts handle rotation into controlled vertical table motion through a threaded nut block and crossed scissor linkage, with the assembly constrained to move through its intended height range without overconstraint or interference. The project also included engineering drawings and a bill of materials for the table mechanism.

Full Assembly

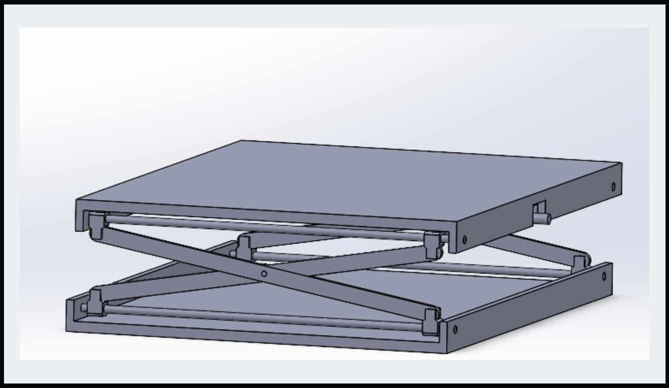


Assembly Drawing

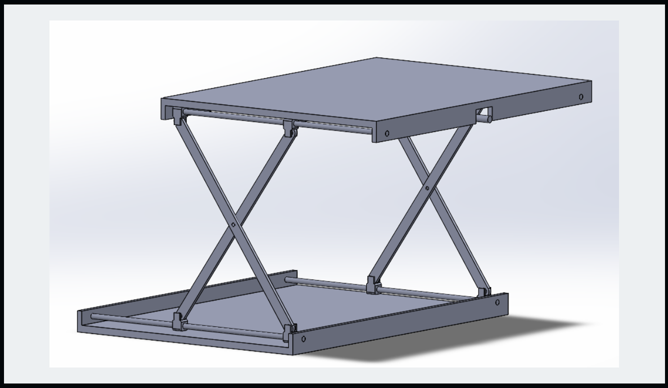


CAD Assembly

Lowered + Raised Positions



LOWERED



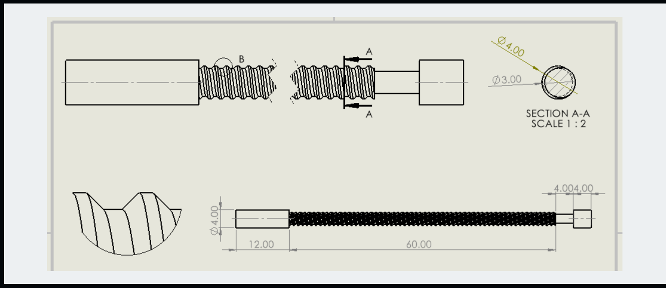
RAISED

Mechanism Breakdown

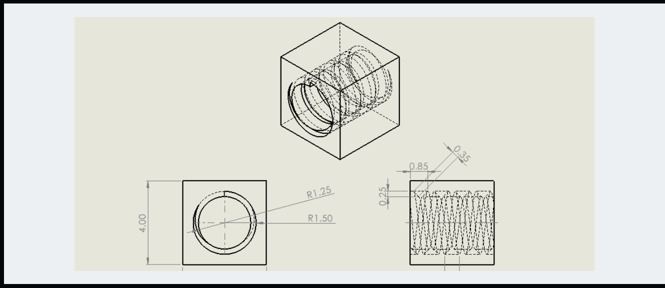
TARGETS	1.5 m by 2.0 m tabletop, approximate 50-140 cm lift range, and 150 kg target payload.
STRUCTURE	Main table frame, tabletop, lower frame, support bars, pivot pins, stop blocks, and scissor lift components.
LIFT	Crossed scissor linkage raises the tabletop as the lower linkage is driven inward.
ACTUATION	Manual lead screw and threaded nut block convert handle rotation into horizontal nut-block translation.
MOTION	SolidWorks screw, concentric, coincident, and limit mates control the motion without overconstraining the assembly.
CLEARANCE	Collapsed-position checks were used because the screw, nut block, rods, frame, and scissor arms are packed closest together.

Lead Screw + Nut Block

I chose a lead screw and threaded nut block so the table could be positioned continuously through the lift range and remain at the selected height without a separate locking mechanism. The trapezoidal thread was chosen for the load-bearing screw interface and to support self-locking at a set height, while the nut block used the matching internal profile required to translate along the screw as the handle turned.



LEAD SCREW DRAWING



NUT BLOCK DRAWING

SolidWorks Motion / Mates

The main design challenge was constraining the table to the required 50-140 cm height range within the given dimensions while still allowing it to stop at any setpoint. I fixed the lead screw axially, prevented the nut block from rotating, and coupled its translation to the scissor linkage so handle rotation remained the single controlled input to the lift.

The mate structure was built around those physical constraints. Concentric and coincident mates defined the pivots and contact geometry, the screw mate converted screw rotation into nut-block travel, and limit mates set the upper and lower bounds of the table motion.